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// Graph Traversal for Campus Navigation Using BFS and  DFS.

#include<stdio.h>
int g[10][10], visited[10], n;
void DFS(int v)
{
 int i;
 printf("%d ", v);
 visited[v] = 1;
 for(i = 0; i < n; i++)
 if(g[v][i]&&!visited[i])
 DFS(i);
}
void BFS(int start)
{
 int q[10], front = 0, rear = 0, i, v;
 for(i = 0; i < n; i++)
          visited[i] = 0;
 q[rear++] = start;
 visited[start] = 1;
 while(front < rear)
 {
 v = q[front++];

printf("%d ", v);
 for(i = 0; i < n; i++)
 {
 if(g[v][i]&&!visited[i])
 {
 q[rear++] = i;
 visited[i] = 1;
 }
 }
 }
}
```

```
int main()
{
 int e, i, j, u, v, start;
 printf("Enter number of Buildings (max 10): ");  scanf("%d", &n);
 if(n < 1 || n > 10)
 return 0;
 printf("Enter number of Paths: ");
 scanf("%d", &e);
 printf("Enter Actual path:\n");
 for(i = 0; i < e; i++)
 {
 scanf("%d%d",&u,&v);
g[u][v] = g[v][u] = 1;
 }


printf("\nAdjacency Matrix:\n");
 for(i = 0; i < n; i++)
 {
 for(j = 0; j < n; j++)
 printf("%d ", g[i][j]);
 printf("\n");
 }
 printf("Enter starting Buildings: ");
 scanf("%d", &start);
 for(i = 0; i < n; i++)
        visited[i] = 0;
  printf("DFS: ");
  DFS(start);
  printf("\nBFS: ");
  BFS(start);
  return 0;
}
```

Output :

```
Enter number of Buildings (max 10): 5
Enter number of Paths: 6
```

```
Enter Actual path:
0 1
0 2
0 3
1 3
3 4
2 4
Adjacency Matrix:
0 1 1 1 0
1 0 0 1 0
1 0 0 0 1
1 1 0 0 1
0 0 1 1 0
Enter starting Buildings: 0
DFS: 0 1 3 4 2
BFS: 0 1 2 3 4
Process returned 0 (0x0)   execution time : 32.475 s Press any key to continue.
```